Design and Implementation of EchoSignRealtime: A Real-Time Wearable Sign Language Recognition System

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***Abstract*—EchoSignRealtime is a wearable sign language recognition system designed to translate hand gestures into real-time textual output using embedded machine learning. The system employs flex sensors and an inertial measurement unit (IMU) interfaced with an ESP32 microcontroller to capture gesture data. A lightweight k-Nearest Neighbors (KNN) classifier performs on-device gesture and sentence-level recognition, while a web-based interface displays results and provides optional speech output. Experimental evaluation demonstrates reliable real-time performance, portability, and cost-effectiveness, making the proposed system suitable for assistive communication applications.**

***Index Terms*—Sign language recognition, wearable sensors, ESP32, embedded machine learning, KNN classifier, assistive technology**

I. Introduction

Communication between hearing-impaired individuals and the general population remains a significant social challenge. Traditional sign language recognition systems rely heavily on vision-based methods, which suffer from limitations such as sensitivity to lighting, background noise, camera positioning, and high computational cost.

EchoSignRealtime addresses these limitations by utilizing a wearable sensor-based approach that captures finger and hand movements directly. This project aims to design and implement a real-time embedded system capable of recognizing hand gestures and short sentences using on-device machine learning and displaying the output through a web-based interface.

II. Methodology

A. System Design

The system consists of sensor data acquisition, preprocessing, on-device classification, and real-time visualization.

B. Components and Specifications

1) Active Components:

- ESP32 Microcontroller
- Flex Sensors
- Inertial Measurement Unit (IMU)
- Power Supply Module
- Wireless Communication Interface

2) Signal Specifications:

- Sensor voltage range: 0–3.3 V
- Sampling frequency: 50–100 Hz
- Communication protocol: Wi-Fi / Serial

3) Test Equipment: Digital multimeter, laptop, USB programmer, regulated power supply.

C. Circuit Design

1) Sensor Interface Circuit: Flex sensors and IMU are interfaced with the ESP32 through analog and digital input channels.

2) ESP32 Processing Circuit:

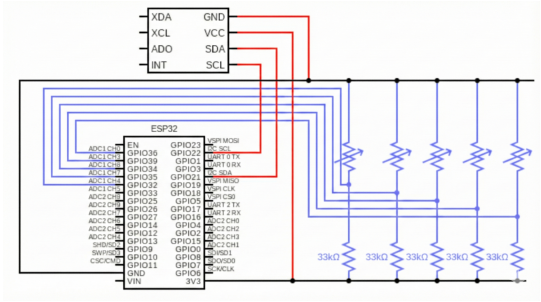


Fig. 1: Block Diagram of Overall System Architecture of EchoSignRealtime

D. Implementation Details

1) Simulation Setup and Model Training: The machine learning model was trained using a comprehensive custom dataset collected from the wearable glove sensors. Raw sensor values were recorded in text files corresponding to individual hand gestures (A–Z and Rest) and common conversational

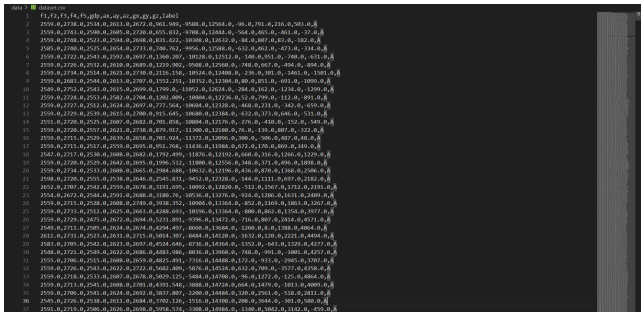
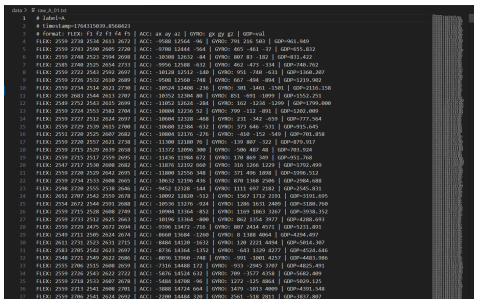


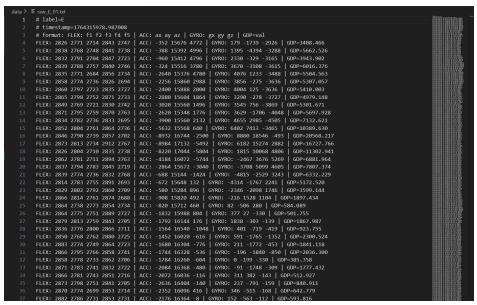
Fig. 2: Model training environment.

sentences. Each file contains time-series data of flex sensor and IMU readings captured during real-time gesture execution.

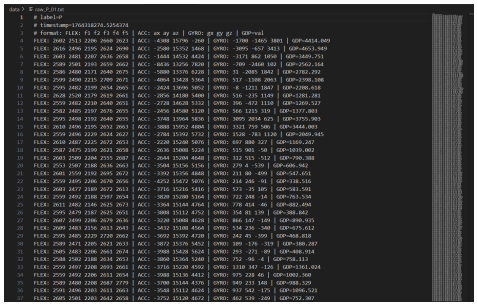
Gesture Dataset Collection:



(a) Gesture A



(b) Gesture E



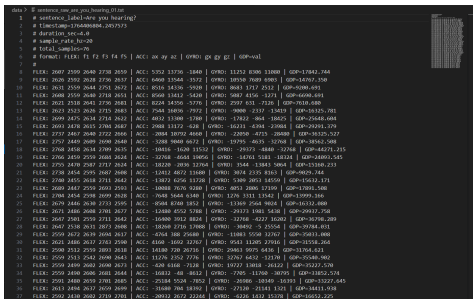
(c) Gesture P

Fig. 3: Sample hand gesture dataset collection process.

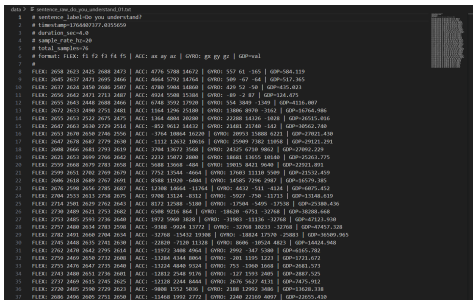
Single-hand alphabet gestures were recorded multiple times to capture natural variations. The dataset includes files such as:

- `raw_A_01.txt`, `raw_A_02.txt`
- `raw_B_01.txt`, `raw_B_02.txt`
- `raw_C_01.txt`, `raw_C_02.txt`
- `raw_D_01.txt`, `raw_E_01.txt`, `raw_F_01.txt`
- `raw_G_01.txt`, `raw_G_02.txt`
- `raw_H_01.txt` to `raw_Y_01.txt`
- `raw_Rest_01.txt`

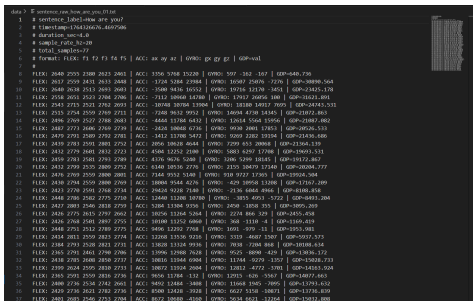
Sentence Dataset Collection:



(a) Gesture [Are you hearing?]



(b) Gesture [Do you understand?]



(c) Gesture [How are you?]

Fig. 4: Continuous sentence gesture recording process.

Continuous gesture sequences were recorded to form natural language expressions. Each sentence was captured ten times to ensure consistency and robustness. The dataset includes sentences such as:

- `are_you_hearing`
- `do_you_understand`
- `do_you_want_help`
- `how_are_you`
- `how_can_i_help_you`
- `no_i_dont_understand`

- see_you_later
- thank_you
- what_time_is_it
- whats_up
- where_do_you_live
- rest

with files such as sentence_raw_how_are_you_01.txt to sentence_raw_how_are_you_10.txt, similarly for all listed sentences.

Training Procedure:

The raw text files were processed using Python-based preprocessing scripts that segmented the data into fixed time windows, normalized sensor values, and extracted relevant statistical features. The processed dataset was used to train two separate k-Nearest Neighbors (KNN) classifiers:

- A gesture-level classifier for alphabet recognition
- A sentence-level classifier for continuous sentence prediction

The trained model parameters were exported as header files and embedded into the ESP32 firmware for real-time on-device inference.

2) Practical Implementation: The ESP32 continuously reads sensor data, performs inference, and transmits recognized output to a web interface.

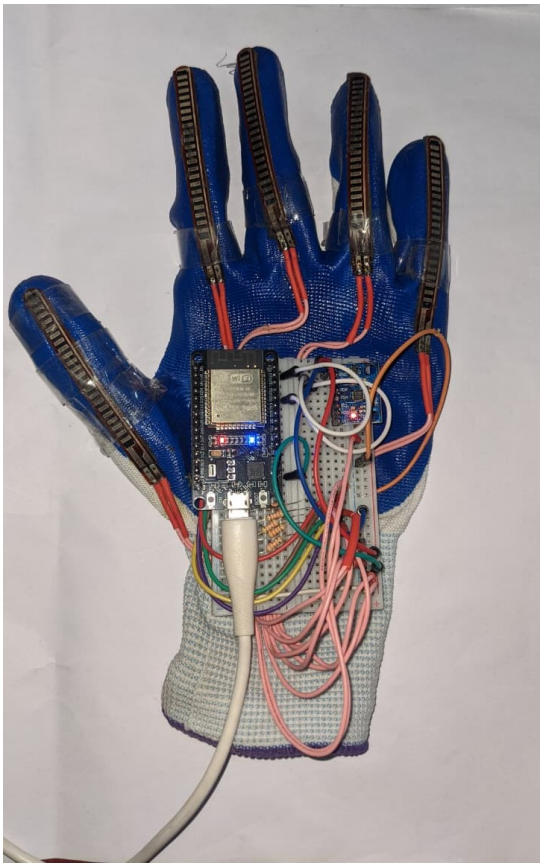


Fig. 5: Practical Hardware Implementation

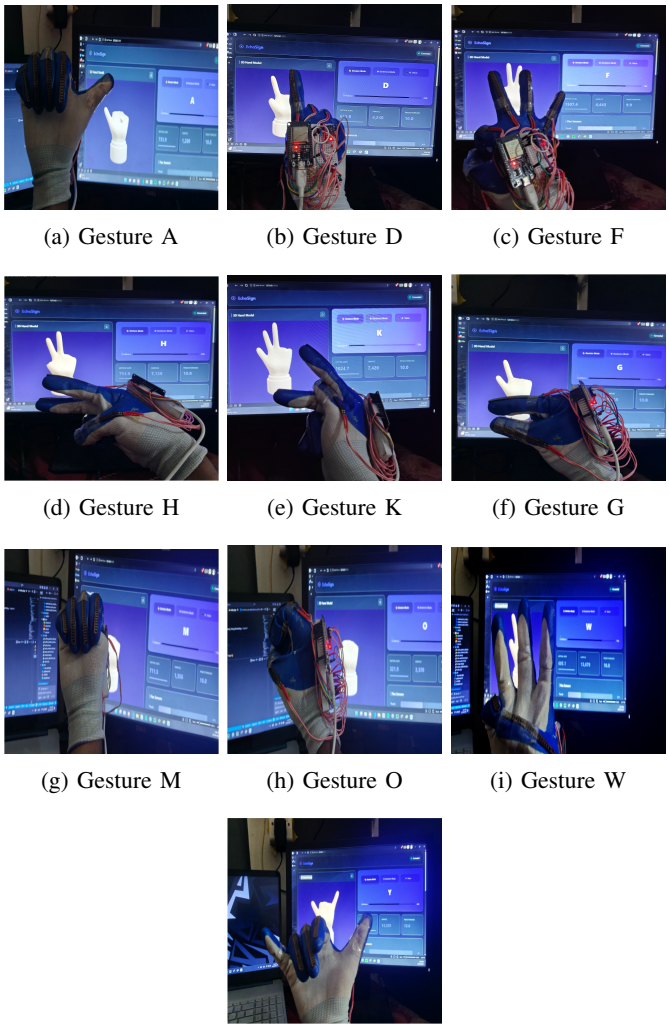
III. Results and Analysis

A. Input Signal Configuration

The input signals were configured consistently for both simulation and practical usage. The gesture recognition system uses real-time flex sensor and IMU data captured from the glove. The sampling rate is 50–100 Hz, and the ESP32 processes the sensor readings to classify gestures and sentences.

B. Practical Implementation Results

The practical implementation was evaluated under real-time conditions using the glove-based system. Photographic documentation and screen captures were obtained to systematically illustrate the recognition performance for individual gestures and continuous sentences.



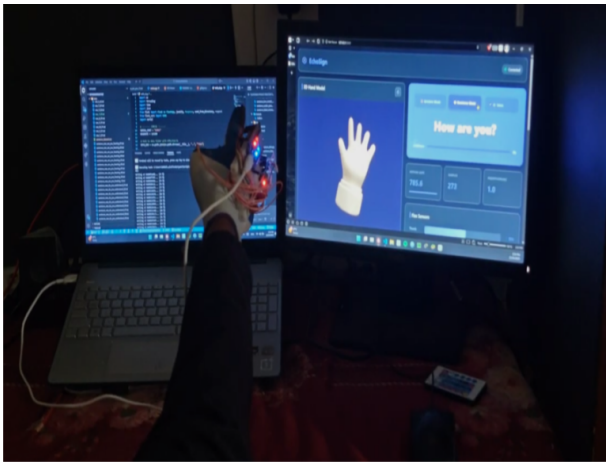
(a) Gesture A (b) Gesture D (c) Gesture F

(d) Gesture H (e) Gesture K (f) Gesture G

(g) Gesture M (h) Gesture O (i) Gesture W

(j) Gesture Y

Fig. 6: Practical gesture recognition results for letters (A–J).



(a) Sentence: How are you??



(b) Sentence: Do you understand?



(c) Sentence: How can I help you?

Fig. 7: Practical sentence recognition results

C. Comparative Analysis

A comparison between simulation results and practical measurements revealed the following observations:

- The gesture recognition accuracy in practical experiments slightly decreased due to sensor noise and variations in hand movement speed.
- Sentence recognition accuracy was affected by cumulative errors from individual gestures, emphasizing the need for robust filtering and preprocessing.
- Overall, the ESP32 successfully processed and recognized both letters and sentences in real-time, demonstrating the feasibility of the system.

D. Performance Limitations

The following limitations were observed during practical implementation:

- Sensor drift and occasional misclassification due to inconsistent hand gestures.
- Limited processing speed on ESP32 affects maximum achievable sampling rate.
- Limited dataset size for sentence sequences affects classifier generalization.
- Environmental lighting and hand placement slightly affect IMU readings.

E. System Limitations and Enhancement Strategies

- Increase training dataset size for both letters and sentences to improve robustness.
- Use advanced preprocessing techniques such as dynamic time warping (DTW) or neural networks for better recognition.
- Optimize ESP32 firmware to reduce latency and allow faster sampling.
- Consider high-precision sensors or additional IMU axes to reduce noise.
- Implement visual feedback on the device for real-time correction of misclassifications.

IV. Conclusion

EchoSignRealtime successfully demonstrates a practical real-time sign language recognition system using wearable sensors and embedded machine learning. The project highlights the effectiveness of on-device inference for assistive communication applications.

References

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[2] R. Yang et al., "Wearable Sensor-Based Sign Language Recognition," IEEE Sensors Journal, 2021.

V. Individual Contributions

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